

Definition

A pushdown automaton is a 6-tuple
 $(Q, \Sigma, \Gamma, \delta, q_0, F)$

- Q is the set of states
- Σ is the input alphabet
- Γ is the stack alphabet
- δ is the transition function

$$\delta: Q \times \Sigma_{\epsilon} \times \Gamma_{\epsilon} \rightarrow P(Q \times \Gamma_{\epsilon})$$

$$\Sigma_{\epsilon} = \Sigma \cup \{\epsilon\} \quad \Gamma_{\epsilon} = \Gamma \cup \{\epsilon\}$$

- q_0 is the start state
- $F \subseteq Q$ is the set of accept states.

A pushdown automaton $M = (Q, \Sigma, \Gamma, \delta, q_0, F)$ accepts input w if w can be written as $w = w_1 w_2 w_3 \dots w_m$, where $w_i \in \Sigma_\epsilon$ and sequence of states $r_0, r_1, r_2, \dots, r_m \in Q$ and strings $s_0, s_1, \dots, s_m \in \Gamma^*$ exist that satisfy the following three conditions.

The strings s_i represent the sequence of stack contents that M has on the accepting branch of computation.

1. $r_0 = q_0$ and $s_0 = \epsilon$. (This condition signifies that M starts properly, in the start state and with an empty stack.)

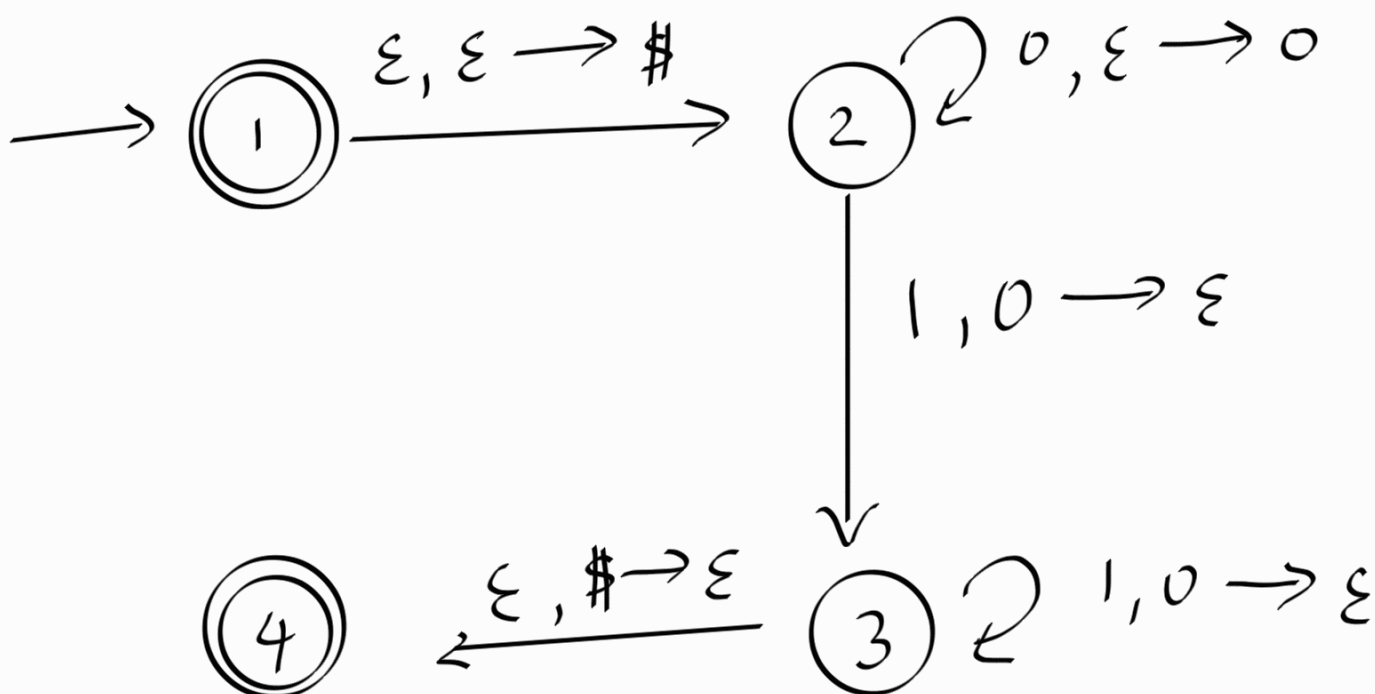
2. For $i = 0, \dots, m-1$, we have $(r_{i+1}, b) \in \delta(r_i, w_{i+1}, a)$ where $s_i = at$ and $s_{i+1} = bt$ for some $a, b \in \Gamma_\epsilon$ and $t \in \Gamma^*$. This condition states that M moves properly according to the state, stack, and next input symbol.

3. $r_m \in F$. This condition states that an accept state occurs at the input end.

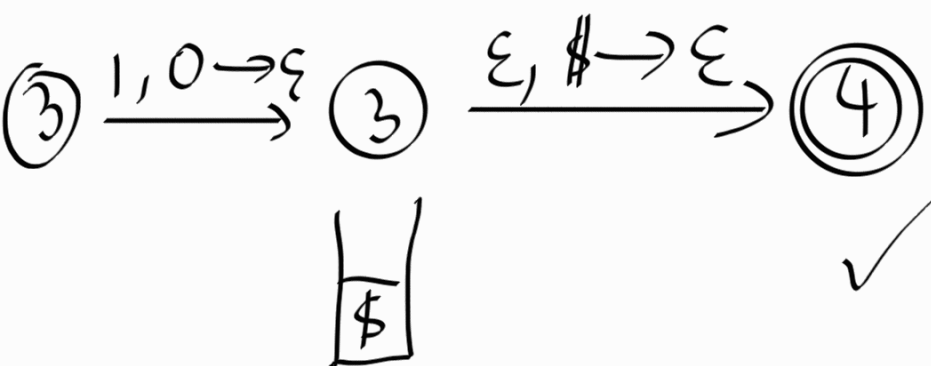
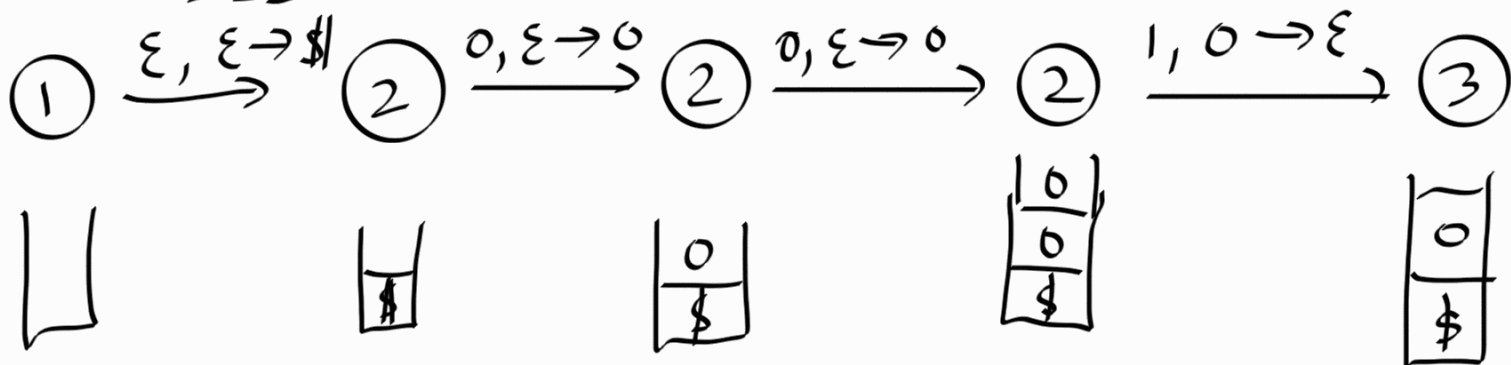
Example of PDA that recognizes $\{0^n 1^n \mid n \geq 0\}$

ε 01 0011
 $\checkmark$ $\checkmark$ $\checkmark$

$$\Gamma = \Sigma_{\varepsilon} \cup \{\#\}$$

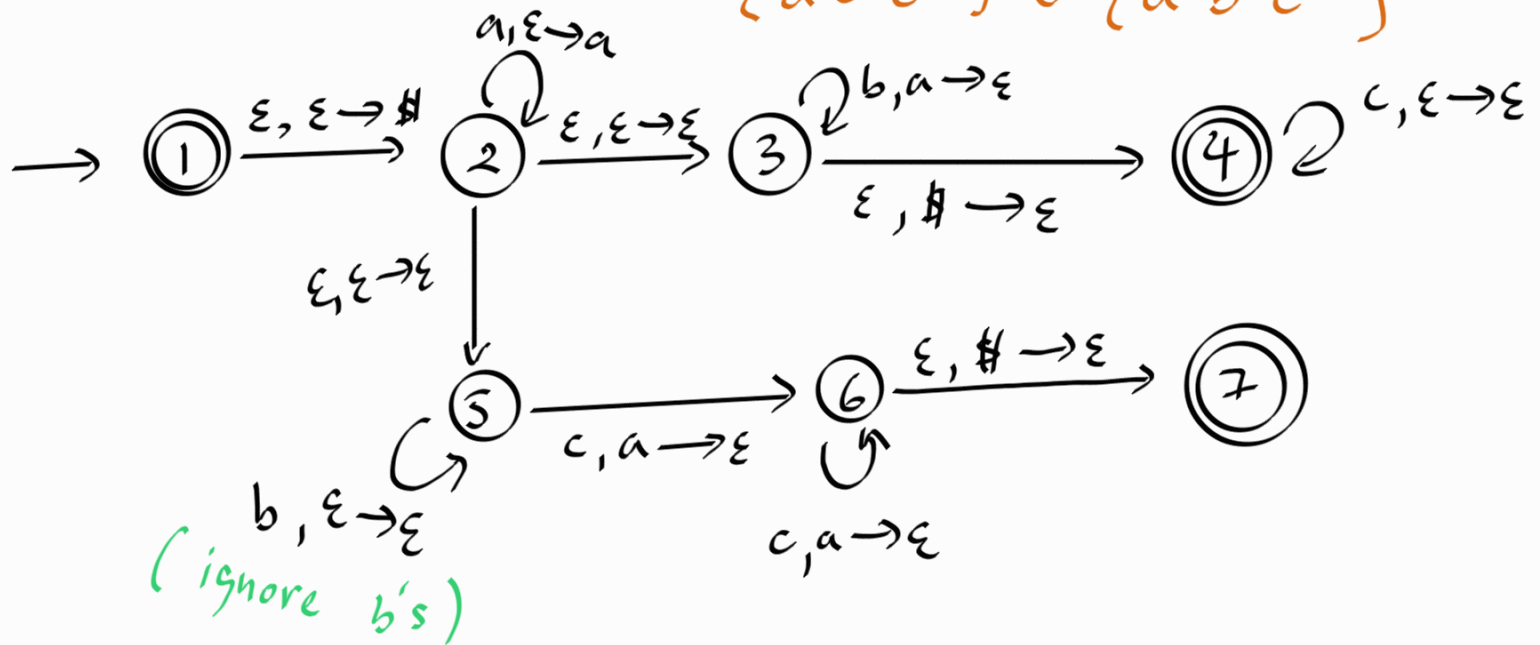


$S = \underline{0011}$



Ex 02: PDA for $A = \{ a^i b^j c^k \mid i, j, k \geq 0, i=j \text{ or } i=k \}$

$\downarrow$
 $\{ a^i b^j c^k \} \cup \{ a^i b^j c^i \}$

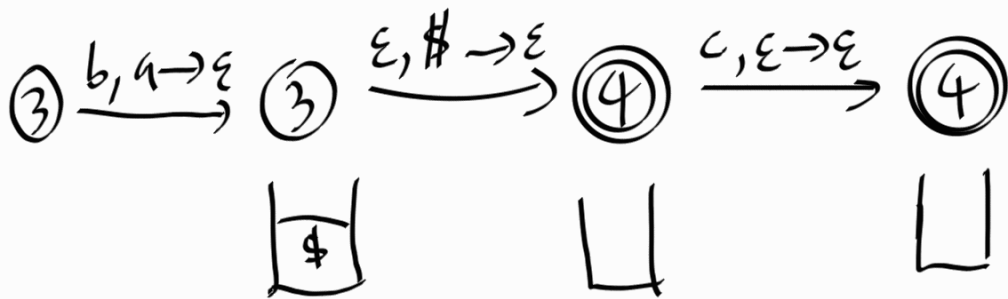
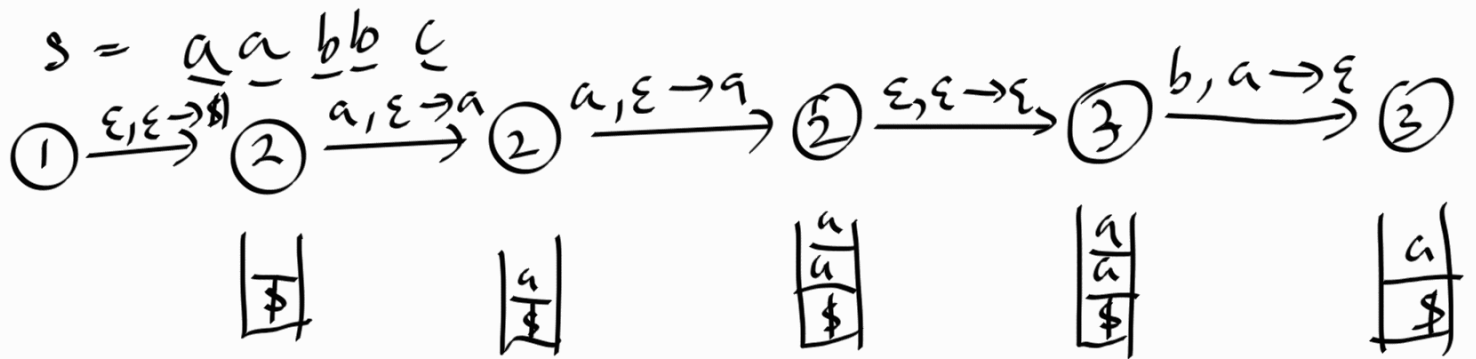
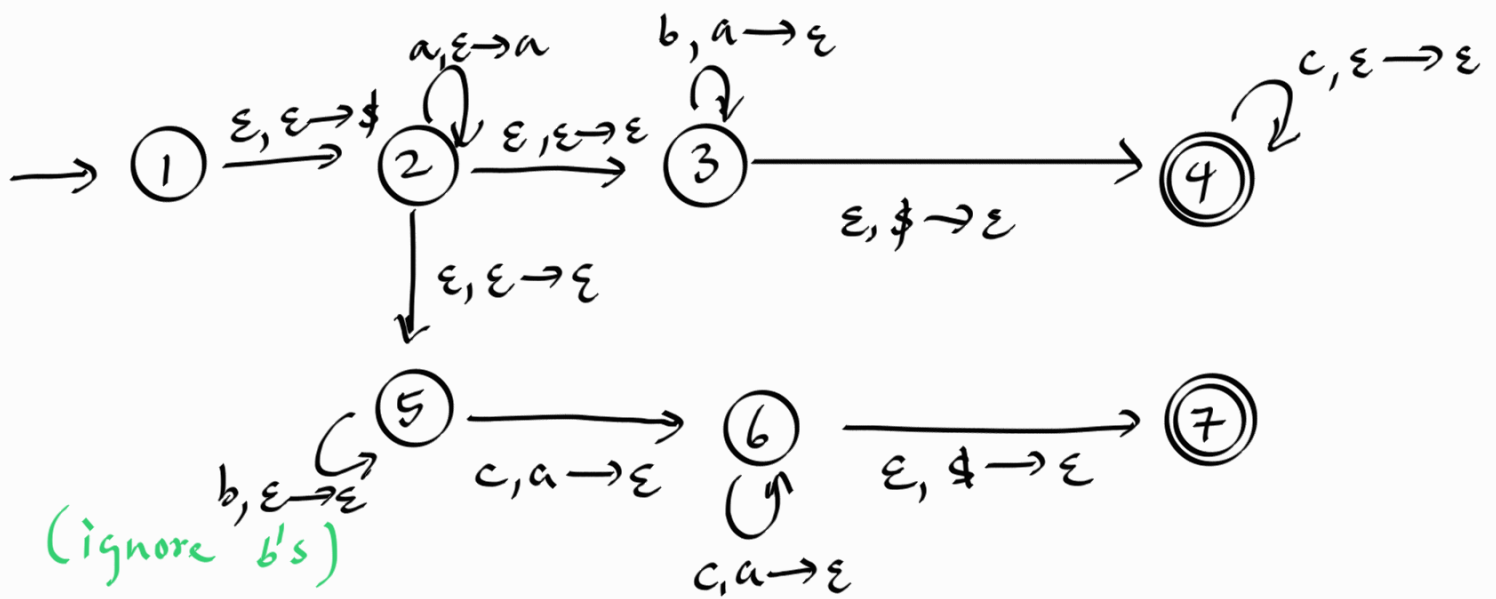


$a, b \rightarrow c$

If a is ϵ : you are reading nothing

If b is ϵ : pop nothing from the stack

If c is ϵ : push nothing to the stack



Ex 3: PDA for $B = \{ ww^R \mid w \in \{0,1\}^+ \}$
 w^R means reverse string of w

$w = 01011$ $w^R = 11010$

$s = 0101111010 \in B$

